

OPENER-Legal: A Frugal Open-World NER Pipeline for the Legal Domain

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Abstract

Open-world named entity recognition (NER) must assign entity types that are not fixed at training time, a setting that is pervasive in the legal domain: new statutes, parties, instruments, courts and jurisdictions appear continually, and re-annotating and retraining for each is costly. We ask whether **OPENER**, a frugal open-world NER pipeline assembled entirely from off-the-shelf pretrained components and trained only on general-domain data, *transfers* to legal NER *without any legal-domain retraining*. We reuse its contrastively fine-tuned, Matryoshka-truncatable entity embedder unchanged (released on the Hugging Face Hub) behind a frozen open-world detector, and evaluate two lightweight typing heads on N **legal NER datasets** (LANGUAGES): **OPENER-Sup**, a balanced linear probe fit on the target labels, and **OPENER-ZS**, a label-free head combining label-name prototypes, a transductive refinement and a fusion with the detector’s own zero-shot predictions. Following the general-domain protocol, we report quality (Adjusted Mutual Information) jointly with inference latency and energy. Our results characterise how far a general-purpose, energy-frugal open-world pipeline carries over to specialised legal schemas, and where domain adaptation would still help.

Index Terms

legal NLP, named entity recognition, open-world NER, zero-shot, contrastive embeddings, frugal AI

I. INTRODUCTION

Legal text is a demanding setting for named entity recognition (NER). Entity inventories are open and shifting (new statutes, case citations, parties, courts, contractual instruments, jurisdictions), documents are long and formulaic, and annotation requires expert effort, so re-annotating and retraining a closed-set tagger for every new type or sub-domain is rarely practical. This is precisely the regime that *open-world* NER targets: recognising and typing entities whose categories were not fixed at training time.

A natural question for practitioners with modest compute is whether an open-world NER system trained on *general-domain* data can simply be *reused* on legal text, without any legal-domain retraining. We study this with **OPENER**, a frugal open-world pipeline assembled from off-the-shelf pretrained components (a frozen detector and a contrastively fine-tuned Matryoshka embedder), which on a general-domain benchmark is competitive in accuracy while staying one to two orders of magnitude cheaper in latency and energy than trained-encoder and generative baselines.

We take **OPENER** exactly as released and apply it to legal NER, reusing its entity embedder from the Hugging Face Hub unchanged. We evaluate two operating points: **OPENER-Sup**, a balanced linear probe fit on the legal target labels, and **OPENER-ZS**, a label-free head (label-name prototypes, transductive refinement, detector fusion). Our contributions are:

- A **transfer study** of a frugal, general-domain open-world NER pipeline to the legal domain, with *no* legal-domain retraining of any component.
- An evaluation on N **legal NER datasets** (LANGUAGES) along three axes at once – quality (AMI), latency and energy – for both a supervised and a zero-shot typing head.
- An analysis of **where the transfer holds and where it breaks** on specialised legal schemas, indicating when light domain adaptation would pay off.

II. RELATED WORK

A. Open-world and zero-shot NER

Open-world NER recognises entity types not fixed at training time. GLiNER [1] scores spans against candidate type names with a bidirectional encoder; GNER [2] casts NER as generation; OWNER [3] trains a dedicated entity encoder and clusters mentions into dynamically named types. Zero-shot typing via type-name semantics has also been studied [4]. **OPENER** differs by reusing a pretrained embedding rather than training an encoder; here we ask how that reuse transfers to the legal domain.

B. Entity representations and contrastive learning

OPENER builds on pretrained sentence embedders [5], [6] with Matryoshka truncation [7], sharpened by a triplet objective [8]; the typing heads use a balanced linear classifier [9] and AMI [10] as the primary metric.

C. Legal NLP and legal NER

Legal NER has its own datasets and domain-specific schemas; we evaluate transfer to them and position OPENER against domain baselines (section IV).

D. Energy-aware evaluation

Reporting efficiency alongside accuracy is increasingly advocated [11]–[13]; we adopt the three-axis (quality/latency/energy) view of the general-domain system, measured with CodeCarbon [14].

III. METHOD

We apply **OPENER** *as is* to the legal domain: the pipeline, the detector and the entity embedder are reused unchanged from the general-domain system, and only the lightweight typing head is fit (or run zero-shot) on the legal targets. No component is retrained on legal data, which lets us measure how well a frugal, general-purpose open-world pipeline *transfers* to legal NER.

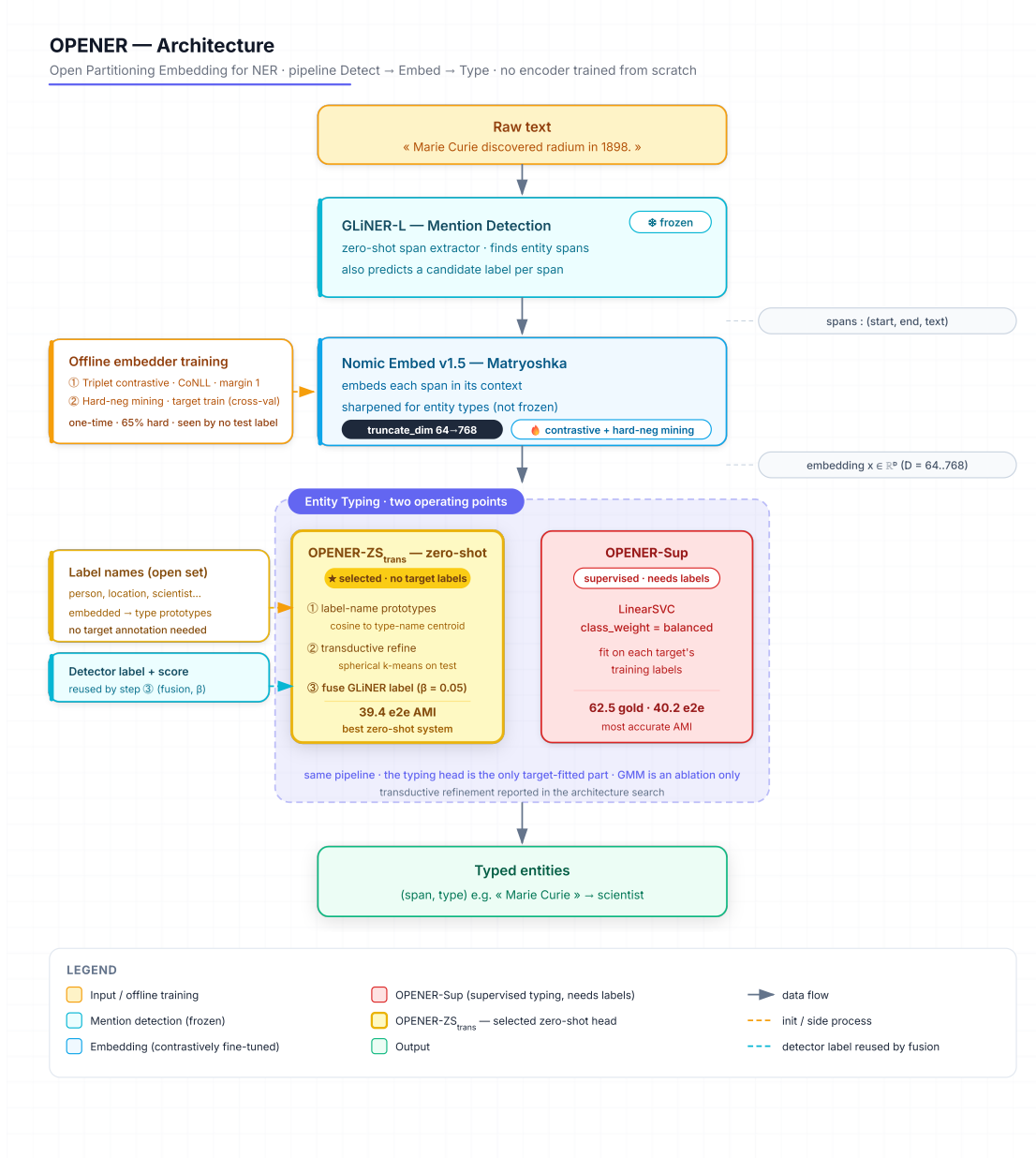


Fig. 1. The OPENER pipeline (Detect → Embed → Type), reused unchanged for legal NER: a frozen GLiNER detector proposes spans, a contrastively fine-tuned Matryoshka embedder maps each span in context to a vector, and a typing head assigns a type. Two operating points: **OPENER-Sup** (supervised linear probe) and **OPENER-ZS** (zero-shot: label-name prototypes, transductive refinement, detector fusion). The embedder is reused from the general-domain model; nothing is retrained on legal data.

A. Mention detection and embedding

Spans are proposed by a frozen **GLiNER-L** detector [1] and embedded *in context* by **Nomic v1.5** [5], a Matryoshka-truncatable sentence embedder [7] that we reuse from the general-domain OPENER, where it was sharpened by a triplet contrastive step and an error-driven hard-negative mining pass. The embedder is released on the Hugging Face Hub and loaded as-is; no legal-domain fine-tuning is performed.

B. Entity typing

a) *OPENER-Sup*: A balanced linear classifier (LinearSVC, `class_weight=balanced` [9]) is fit on the legal target’s training labels. This is the supervised upper bound; it needs target labels.

b) *OPENER-ZS*: The zero-shot head uses no target labels: each legal type t is represented by the embedding of its name (a label-name prototype); a mention is assigned the nearest prototype by cosine similarity, refined by a transductive spherical k -means over the unlabelled test mentions, and *fused* with the detector’s own zero-shot label,

$$s_t(i) = \cos(e_i, p_t) + \beta g_i \mathbb{1}[t = \ell_{\text{GLiNER}}(i)], \quad \beta=0.05. \quad (1)$$

C. Evaluation

We report both *typing-on-gold* (perfect detection) and *end-to-end* protocols, aligning predicted and gold mentions by character offsets with sentinel labels for detection errors, following the general-domain setup. Quality is the Adjusted Mutual Information (AMI) [10], complemented by macro- F_1 ; we also report per-sentence latency (p50) and energy (Wh, gCO₂eq via CodeCarbon [14]).

IV. EXPERIMENTS

A. Datasets

We evaluate on N legal NER datasets spanning LANGUAGES and SUB-DOMAINS (e.g. case law, contracts, legislation). Table I summarises them and their entity-type inventories.

TABLE I
LEGAL NER DATASETS (PLACEHOLDER). “TEST” IS THE NUMBER OF TEST SENTENCES (CAPPED AT 1000); EACH TYPE IS FOLLOWED BY ITS SHARE (%) OF THE TEST MENTIONS.

Dataset	Lang	Test	Entity types (share %)
<i>TODO: fill once datasets are provided</i>			

B. Setup

Embedder, detector and heads are exactly those of the general-domain OPENER (section III); the entity embedder is loaded from the Hugging Face Hub and is *not* retrained on legal data. Single fixed seed; each test split capped at 1000 sentences. Baselines:

C. Results

Table II reports AMI (gold and end-to-end) per legal dataset for OPENER-Sup and OPENER-ZS, alongside latency and energy.

TABLE II
LEGAL NER RESULTS (PLACEHOLDER): AMI ($\times 100$, $\uparrow$), GOLD AND END-TO-END.

System	AMI gold	AMI e2e
GLiNER-L (zero-shot)	–	–
OPENER-ZS	–	–
OPENER-Sup	–	–

D. Analysis

V. CONCLUSION

We studied whether **OPENER**, a frugal open-world NER pipeline assembled from off-the-shelf pretrained parts and trained only on general-domain data, transfers to legal NER without any legal-domain retraining. Reusing its entity embedder unchanged, we evaluated a supervised linear probe (OPENER-Sup) and a label-free head (OPENER-ZS: prototypes, transductive refinement, detector fusion) on N legal datasets along quality, latency and energy. Our findings indicate [MAIN TAKEAWAY]; the residual gap on specialised legal schemas points to light domain adaptation (legal-domain contrastive fine-tuning of the same embedder) as the natural next step, while keeping the frugal, decoupled detect-then-type design intact.

a) *Limitations*: A single random seed; capped test splits; the entity embedder is reused without legal-domain adaptation, and detection relies on a general-domain open-world detector whose recall on specialised legal spans may cap end-to-end quality.

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